

2. kontrolna naloga

3. A, 3. 12. 2024

Ime in priimek: MITJA ŠEVERKAR

dosežene točke	možne točke	odstotki	ocena
30	36	83	4

ČAS PISANJA: 45 minut

$$\begin{array}{r} 503 \\ - 360 \\ \hline 143 \end{array}$$

$$\begin{array}{r} 360 \\ - 233 \\ \hline 127 \end{array}$$

$$\begin{array}{r} 160 \\ 360 \\ 540 \\ 720 \end{array}$$

1. Natančno izračunaj vrednost izraza: $\sin 503^\circ + \tan(-480^\circ) + \cos(-233^\circ)$. [6t] 6

$$\begin{aligned} \sin 503^\circ + \tan(-480^\circ) + \cos(-233^\circ) &= \sin 143^\circ + \tan 60^\circ + \cos 127^\circ = \\ &= \sin(180^\circ - 37^\circ) + \sqrt{3} + \cos(180^\circ - 53^\circ) = \sin 37^\circ + \sqrt{3} - \cos 53^\circ = \\ &= \sin 37^\circ - \sin 37^\circ + \sqrt{3} = \underline{\underline{\sqrt{3}}} \end{aligned}$$

$$\tan 60^\circ = \frac{\sin 60^\circ}{\cos 60^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

2. Dokaži: $\sin^2(x + \frac{\pi}{6}) + \cos^2(x - \frac{\pi}{6}) = 1 + \frac{\sqrt{3}}{2} \sin(2x)$.

[7t] 7

$$\begin{aligned} L: \sin^2(x + \frac{\pi}{6}) + \cos^2(x - \frac{\pi}{6}) &= (\sin x \cdot \cos 30^\circ + \cos x \cdot \sin 30^\circ)^2 + (\cos x \cdot \cos 30^\circ - \sin x \cdot \sin 30^\circ)^2 \\ &= (\frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x)^2 + (\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x)^2 = \frac{3}{4} \sin^2 x + \frac{\sqrt{3}}{2} \sin x \cos x + \frac{1}{4} \cos^2 x \\ &\quad + \frac{3}{4} \cos^2 x + \frac{\sqrt{3}}{2} \cos x \sin x + \frac{1}{4} \sin^2 x = \sin^2 x + \cos^2 x + \sqrt{3} \sin x \cos x \\ &= 1 + \sqrt{3} \sin x \cos x = 1 + 2 \cdot \frac{\sqrt{3}}{2} \sin x \cos x = \underline{\underline{1 + \frac{\sqrt{3}}{2} \sin(2x)}} = D \end{aligned}$$

$$D: 1 + \frac{\sqrt{3}}{2} \sin(2x)$$

3. Izračunaj natančno vrednost izraza $\cos^2 \frac{\alpha}{2} + \tan(\beta - \frac{\pi}{4})$, če je $\tan \alpha = 2\sqrt{2}$ ($0^\circ < \alpha < 90^\circ$) in $\sin \beta = -\frac{3}{5}$ ($180^\circ < \beta < 270^\circ$).

[8t] 8

$$\begin{aligned} \cos^2 \frac{\alpha}{2} + \tan(\beta - \frac{\pi}{4}) &= \cos^2 \frac{\alpha}{2} + \tan(\beta - 45^\circ) \\ &= \left(\pm \sqrt{\frac{1+\cos \alpha}{2}} \right)^2 + \frac{\tan \beta - \tan 45^\circ}{1 + \tan \beta \cdot \tan 45^\circ} \\ &= \frac{1+\cos \alpha}{2} + \frac{\tan \beta - 1}{1 + \tan \beta} = \frac{1+\cos \alpha}{2} + \frac{\tan \beta - 1}{1 + \tan \beta} \\ &= \frac{1+\cos \alpha}{2} + \frac{\frac{3}{4} - 1}{1 + \frac{3}{4}} = \frac{1+\frac{1}{3}}{2} + \frac{\frac{3}{4} - 1}{1 + \frac{3}{4}} = \frac{\frac{4}{3}}{2} + \frac{-\frac{1}{4}}{\frac{7}{4}} = \frac{2}{3} - \frac{1}{7} = \frac{14}{21} - \frac{3}{21} = \frac{11}{21} \end{aligned}$$

$$\begin{aligned} 1 + \tan^2 \alpha &= \frac{1}{\cos^2 \alpha} \\ \cos \alpha &= \sqrt{\frac{1}{1 + \tan^2 \alpha}} = \sqrt{\frac{1}{1 + 8}} = \sqrt{\frac{1}{9}} = \frac{1}{3} \\ \cos \beta &= -\sqrt{1 - \sin^2 \beta} = -\sqrt{1 - \frac{9}{25}} = -\sqrt{\frac{16}{25}} = -\frac{4}{5} \\ \tan \beta &= \frac{\sin \beta}{\cos \beta} = \frac{-\frac{3}{5}}{-\frac{4}{5}} = \frac{3}{4} \end{aligned}$$

4. Dokaži, da je $\sin 48^\circ \cdot \sin 12^\circ \cdot \cos 18^\circ = \frac{1}{2} \cos 54^\circ$.

$\frac{1}{2}$

[5t] 2

$$\begin{aligned} \sin 48^\circ \cdot \sin 12^\circ \cdot \cos 18^\circ &= -\frac{1}{2} (\cos(60^\circ) - \cos(36^\circ)) \cdot \cos 18^\circ \\ &= -\frac{1}{2} \cdot \frac{1}{2} \cdot \cos 18^\circ + \frac{1}{2} \cos 18^\circ \cdot \cos 36^\circ = \frac{1}{4} \cos 18^\circ + \frac{1}{2} \cos 18^\circ \cdot \cos 36^\circ \\ &= -\frac{1}{4} \cos 18^\circ + \frac{1}{2} \cos 18^\circ \cdot (\cos^2 18^\circ - \sin^2 18^\circ) \\ &= -\frac{1}{4} \cos 18^\circ + \frac{1}{2} \cos 18^\circ (2\cos^2 18^\circ - 1) \\ &= -\frac{3}{4} \cos 18^\circ + \cos^3 18^\circ = \cos 18^\circ \left(-\frac{3}{4} + \cos^2 18^\circ \right) \\ &= \cos 18^\circ (-\cos^2 30^\circ + \cos^2 18^\circ) \\ &= \cos 18^\circ (\cos^2 18^\circ - \cos^2 30^\circ) \end{aligned}$$

5. Dana je funkcija $f(x) = -3 \sin(\frac{3}{2}x + \frac{3\pi}{4})$.

a) Na intervalu $[-2\pi, 3\pi]$ nariši graf funkcije f .

[4t] 4

ničle:

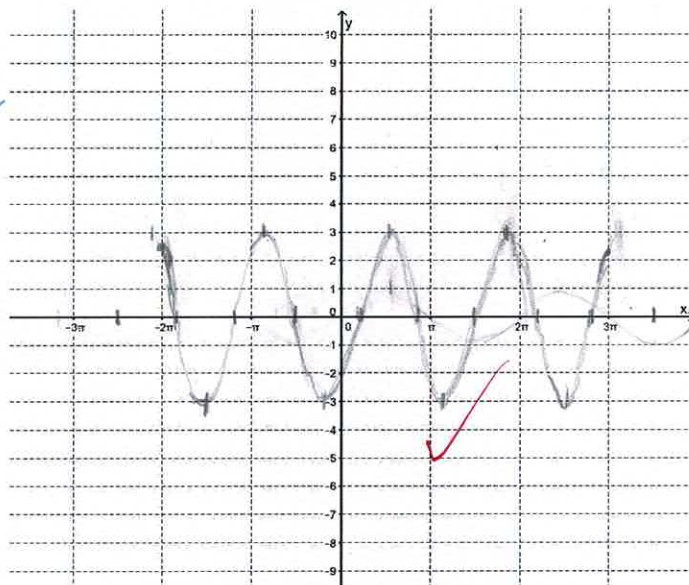
$$0 = -3 \sin(\frac{3}{2}x + \frac{3\pi}{4}) \quad k \in \mathbb{Z}$$

$$0 = \sin(\frac{3}{2}x + \frac{3\pi}{4})$$

$$k\pi = \frac{3}{2}x + \frac{3\pi}{4}$$

$$k\pi - \frac{3\pi}{4} = \frac{3}{2}x$$

$$x = \frac{2}{3}\pi \cdot k - \frac{1}{2}\pi$$



b) Računsko določi maksimume funkcije f .

[4t] 2

$$g(x) = \sin(\frac{3}{2}x + \frac{3}{4}\pi) \quad \mathcal{Z}_g = [-1, 1]$$

$$f(x) = -3g(x) \quad \mathcal{Z}_f = [-3, 3]$$

$$3 = -3 \sin(\frac{3}{2}x + \frac{3\pi}{4})$$

$$-1 = \sin(\frac{3}{2}x + \frac{3\pi}{4})$$

$$2k\pi + \frac{3}{2}\pi = \frac{3}{2}x + \frac{3\pi}{4}$$

$$2k\pi = \frac{3}{2}x$$

$$x = \frac{4}{3}k\pi \quad k \in \mathbb{Z}$$

$$2k\pi + \frac{6}{4}\pi - \frac{3}{4}\pi = \frac{3}{2}x$$

$$\frac{3}{2}x = 2k\pi + \frac{3}{4}\pi$$

$$x = \frac{4}{3}k\pi + \frac{1}{2}\pi \quad k \in \mathbb{Z}$$

f+oče

c) S katerima togima premikoma iz krivulje $y = \sin(2x - \frac{\pi}{4}) + 3$ dobimo krivuljo $y = \cos(2x) - 3$?

[2t] 1

$$y = \sin(2x - \frac{\pi}{4}) + 3 = \cos(\frac{2\pi}{4} + \frac{\pi}{4} - 2x) + 3 = \cos(\frac{3\pi}{4} - 2x) + 3 = \cos(2x - \frac{3\pi}{4}) + 3$$

S premikom v ~~desno~~ ^{levo} za $\frac{3}{8}\pi$ in s premikom za $\frac{6}{8}\pi$ enot na dol (zmanjšuje se y koordinata)

premaknjeno na desno

DODATNA NALOGA: Izračunaj: $\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ}$.

[3t] 

